

TANMAY SHINDE

AI/ML Engineer | GenAI, LLM & Agentic Systems

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SUMMARY

AI/ML engineering student (B.Tech, May 2027) who builds LLM applications and autonomous agents end-to-end and ships them. Software Development Intern at Pixaflip Technologies, where I built Pixa Agent – an open-source AI coding agent for VS Code with 181 automated tests. Prior: deep-learning research internship with IEEE EMBS, a Best Research Paper award, and Top 25 of 600+ teams at a national hackathon. Strong Python and TypeScript engineering discipline with a test-and-review mindset.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

GenAI / LLM / Agents: LLM Agents, Agentic Tool-Calling, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering, Structured/JSON Output, Guardrails, Groq API, OpenAI API, Gemini API, Hugging Face Transformers, FAISS, ChromaDB, Vector Databases, Fine-tuning (LoRA, PEFT)

Machine Learning: Deep Learning, Convolutional Neural Networks (CNN), Ensemble Learning, PyTorch, TensorFlow, scikit-learn, XGBoost, Model Evaluation, Grad-CAM Explainability, NLP (spaCy, distilBERT)

Backend & Infrastructure: FastAPI, Flask, Pydantic, Node.js, Next.js, REST APIs, PostgreSQL, SQLAlchemy, Redis, Docker, GitHub Actions (CI/CD), Linux

Testing & Tools: pytest, Vitest, Git, GitHub, VS Code Extension API

EXPERIENCE

Software Development Intern, Pixaflip Technologies Pvt. Ltd.

Jun 2026 – Present | Pune, India

- Built **Pixa Agent**, an open-source AI coding agent for VS Code, as a core deliverable of this internship – architecture, tests, and security design detailed under Projects.
- Engineering **Acadflip DigGen**, a document-to-courseware pipeline that converts PDFs and slides into fact-checked study modules, decks, and grounded Q&A, using adversarial per-slide validation and multi-provider LLM routing across Groq, OpenAI, and Gemini.

Research Intern, Deep Learning, IEEE EMBS Pune Chapter

Jun 2026 – Jul 2026 | Pune, India

- Built a four-stage PyTorch pipeline for Alzheimer's detection from brain MRI, combining a heterogeneous CNN ensemble, optimization-based weight tuning, and Grad-CAM explainability for auditable predictions.
- Enforced evaluation rigor with subject-wise splits to eliminate data leakage and external validation on a second dataset (ADNI); delivered IEEE-formatted reports under faculty mentorship.

PROJECTS

Pixa Agent – Open-Source AI Coding Agent for VS Code

github.com/WisdomBoost-LLC/PixaAgent

- Built an agent loop that plans multi-step coding tasks, reads the codebase, and edits multiple files, staging every change as an approvable diff before it touches disk.
- Architected a provider-agnostic core around a single ModelProvider interface – any OpenAI-compatible endpoint (Ollama, vLLM, NVIDIA NIM, Groq) – enabling fully offline, vendor-free use.
- Implemented semantic code search (custom chunker, embedding index, vector store) and a security layer with sandboxed commands, secret redaction, and audit logging.
- Maintained 181 automated tests across 26 suites, CI-gated on every push (~7,800 lines of TypeScript, MIT licensed).

Drone Security Analyst Agent

Apr 2026

- Built a video-intelligence agent that answers natural-language questions about drone footage, combining CLIP visual embeddings, a vision-language model, and ChromaDB retrieval with cross-frame entity memory; scored 5/5 on the FlytBase selection benchmark.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning

Expected May 2027

Pimpri Chinchwad University, Pune | CGPA: 8.24 / 10

ACHIEVEMENTS

- Best Research Paper Award, ICCTVB-25** – first-author paper on ensemble ML for renewable energy forecasting.
- Top 25 of 600+ teams**, DP World x BITS Pilani National Hackathon, Hyderabad.
- NVIDIA** – Fundamentals of NLP certification.